

## Multiple Choice

1. **Answer:** C

*Options:* A) True, True; B) False, False; C) True, False; D) False, True

*Note:* Correct option: C - True, False

2. **Answer:** A

*Options:* A) True, True; B) False, False; C) True, False; D) False, True

*Note:* Correct option: A - True, True

3. **Answer:** A

*Options:* A) 4; B) 7; C) 8; D) 12

*Note:* Correct option: A - 4

4. **Answer:** C

*Options:* A) local maximum at P and at Q; B) saddle point at P and at Q; C) local maximum at P and a saddle point at Q; D) local minimum at P and a saddle point at Q

*Note:* Correct option: C - local maximum at P and a saddle point at Q

5. **Answer:** A

*Options:* A) 6; B) 12; C) 30; D) 105

*Note:* Correct option: A - 6

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'False, True'.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: 'True, False'.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* LLM-generated false statement. Correct answer: ' $12\sqrt{3}$ '.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

10. **Answer:** True

*Options:* A) True; B) False

*Note:* LLM-generated true statement based on MCQ.

## Long Form Response

11. **Answer:** Every solution is a scalar multiple of  $(-5, 1, 1, 0)$ .. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** I, II, and III. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

*Note:* Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

*End of Answer Key*